Brendon McGuinness

Curriculum Vitae

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EDUCATION

McGill University

PhD. Quantitative Life Sciences 2025

Advisors: Dr. Frédéric Guichard & Dr. Steph Weber

Thesis title: Trait plasticity and the maintenance of biodiversity

McGill University

BSc. Joint Major Physics & Computer Science 2018

Thesis title: Image analysis tools for investigating topology mediated dynamics in supercoiled DNA

Advisor: Dr. Sabrina Leslie

Graduated on the Dean's Multidisciplinary Undergraduate Research List

PROFESSIONAL APPOINTMENTS

New York University

Postdoctoral Associate 2026-present

Advisor: Dr. Mercedes Pascual

Working on computational models of host–pathogen eco-evolutionary dynamics

PUBLICATIONS

Peer reviewed:

McGuinness, B., Weber, S. C., & Guichard, F. (2025). Resource-use plasticity governs the causal relationship between traits and community structure in model microbial communities. *Proceedings of the National Academy of Sciences*, *122*(29), e2419275122.

Negrín Dastis, J. O., **McGuinness, B.**, Tadiri, C. P., Yargeau, V., & Gonzalez, A. (2024). Connectivity mediates the spatial ecological impacts of a glyphosate-based herbicide in experimental metaecosystems. *Oecologia*, *205*(3), 709-723.

Scott, S., Shaheen, C., **McGuinness, B.**, Metera, K., Kouzine, F., Levens, D., ... & Leslie, S. (2019). Single-molecule visualization of the effects of ionic strength and crowding on structure-mediated interactions in supercoiled DNA molecules. *Nucleic acids research*, *47*(12), 6360-6368.

In revision:

McGuinness, B., Weber, S. C., & Guichard, F. (2026). Resource-use plasticity modulates the resource-diversity relationship in model microbial communities. *Ecology Letters* (accepted)

Preprints:

McGuinness, B., Guichard, F., & Weber, S. C. (2026). The winding road to coexistence: Interdependence of stabilizing and equalizing mechanisms in competing *E. coli* with targeted nutrient transporter gene deletions. *bioRxiv,* 2026-06.

TEACHING EXPERIENCE

Teaching Assistant

Cell and Molecular Biology (undergraduate) – BIOL 112 2025

Ecological Dynamics (undergraduate) – BIOL 308 2022-2023

Foundations of Programming (undergraduate) – COMP 202 2019-2022

Organismal Biology (undergraduate) – BIOL 111 2022

Applied Machine Learning (graduate) – COMP 551 2021

Eukaryotic Cell Biology (graduate) – BIOL 303 2020

WORK & RESEARCH EXPERIENCE

Ph.D. Thesis Student 2019-2025

McGill University: Metabolic plasticity and the maintenance of biodiversity

Supervisors: *Dr. Frédéric Guichard & Dr. Steph Weber*

- Developed consumer-resource models with adaptive trait plasticity to determine how the rate of plasticity shapes the distribution of species relative abundances.
- Explored how trait-abundance feedback interactions can predict shifts in resource-diversity relationships due to transitions in niche-neutral dynamics.
- Built a genome-scale dynamic flux balance analysis (dFBA) pipeline to test modern coexistence theory, showing that stronger coupling of stabilizing and equalizing forces raises the minimum niche separation needed for coexistence; Compared predictions to pairwise competition assays with engineered E. coli strains.

Ph.D. Rotation Student, McGill University 2018-2019

Supervisors: *Dr. Steph Weber* (1)*, Dr. Frédéric Guichard* (2)*, Dr. Jennifer Sunday* (3)

- **Biophysics & Cell Biology (Weber Lab):** Investigated phase-separated protein condensates under confinement using Convex Lens-Induced Confinement (CLiC) microscopy.
- **Theoretical Ecology (Guichard Lab):** Modeled spatially explicit predator-prey networks to examine how ecological mechanisms shape network modularity, local and regional stability, and system responses to fragmentation and perturbation.
- **Machine Learning in Ecology (Sunday Lab):** Applied convolutional neural networks (CNNs) to classify phytoplankton species from flow cytometry images (FlowCam), achieving ~95% classification accuracy. This method enabled species-level resolution in analyzing competitive dynamics for limited essential resources across a temperature gradient.

Research Assistant, McGill University 2017-2018

Supervisor: *Dr. Sabrina Leslie*

- Developed image analysis software to quantify topological changes in DNA and their impact on DNA–DNA interactions. Implemented machine learning algorithms to semi-automate data analysis.

Research Assistant, Health Informatics Group, McGill University 2018

Supervisor: *Dr. John Kildea*

- Student researcher on the Opal app which is designed to give radiation oncology patients access to their medical records, test results, and educational material
- Developed a platform on the Opal app for hospital patients to donate their de-identified medical data to a secure repository for medical research

St. Brigid's Summer Camp 2012-2016

- *Senior Program Director* - Designed and ran daily camp programming, managed logistics, and ensured smooth communication between program staff and operations for charity-run camp.
- *Head Counsellor / Camp Counsellor* - Supervised, trained, and supported teams of counselors; helped resolve conflicts and ensured a safe, inclusive environment for all campers.

PROJECTS

Living Data Project - Garry Oak Ecosystem Restoration Dataset (2025)

Digitized, curated, and standardized historical vegetation-community data from Garry Oak ecosystems in southwestern British Columbia, converting archival thesis records into a reproducible dataset for biodiversity and long-term restoration research. Published through Harvard Dataverse (DOI: 10.7910/DVN/EZJPRO).

Read(the_room) (2023-2025)

Past Contributor – Real-time sentiment mapping platform designed to visualize collective human sentiment. Started as a hackathon project. Developed backend infrastructure and user interface. Focused on anonymized data aggregation, transparency, and user privacy. Website captures and displays trending questions. Emphasizes transparency and community-building in contrast to traditional data-mining platforms. https://readtheroom.site/

Exploring Morality with ChatGPT (2023)

Collaborative project comparing moral-foundation responses from large language models and humans using the Moral Foundations Questionnaire and statistical measures of distributional similarity.

PRESENTATIONS

Selected oral presentations:

Gordon Research Seminar: Unifying Ecology Across Scales (2026) - *The winding road to coexistence: Interdependence of niche and fitness differences in E. coli with targeted resource uptake gene deletions*

Gordon Research Seminar: Stochastic physics in biology (2025) - *Resource-use plasticity mediates the resource-diversity relationship in model microbial communities*

Ecological Society of America Annual Meeting Long Beach (2024) - *Plasticity in resource consumption traits shapes community assembly*

Ecological Society of America Annual Meeting Portland (2023) - *Plasticity in resource consumption traits shape community structure governing the strength of niche and neutral processes in competitive communities*

Workshop sur les Avancées dans la théorie des réseaux écologiques: vers une intégration des relations entre communautés et écosystèmes *(2023) - Higher order interactions and trait plasticity: consequences for community structure*

Centre for Applied Mathematics in Bioscience and Medicine (CAMBAM) Symposium (2023) - *Do species' traits matter? Consequences of trait plasticity on the structure of ecological communities*

Montreal Biophysics Seminar (2022) – *The consequences of metabolic plasticity for coexistence in microbial communities: bridging levels of biological organization*

Poster presentations:

Gordon Research Conference: Unifying Ecology Across Scales (2026) - *The winding road to coexistence: Interdependence of niche and fitness differences in E. coli with targeted resource* uptake gene deletions

Gordon Research Conference: Stochastic physics in biology (2025) - *Resource-use plasticity mediates the resource-diversity relationship in model microbial communities*

Canadian Biophysical Society Annual Meeting Vancouver (2018) - *Investigating Structure-mediated Dynamics and Interactions in Supercoiled DNA*.

OUTREACH

Mentor *- McGill Physics Hackathon (2023)*

R workshop host *- Designed and led R workshop for biology students doing undergraduate research projects (2019)*

Space Explorers program *(volunteer) - Planned various physics activities to bring into elementary school classrooms over the course of the school year* (2018)

WORKING GROUPS AND WORKSHOPS

Santa Fe Institute Complex Systems Summer School (2026)

Ecology of Inter-City Networks - *Developed a network metapopulation model of post-disaster population displacement and mortality, examining how urban connectivity and receiving-city capacity shape vulnerability.*

Exploring Movie Structure Across Time - *Developed temporal character-network methods to quantify and compare changes in movie structure across time.*

Integrated Mathematical Oncology Workshop (2025) – *Steering hepato-metastatic early-onset colorectal cancer into evolutionary traps*

Workshop sur *les Avancées dans la théorie des réseaux écologiques: vers une intégration des relations entre communautés et* écosystèmes (2023) (organizer)

AWARDS

Integrated Mathematical Oncology Workshop Travel Award 2025 ($2000)

Quebec Oceans Travel Award 2025 *($1000)*

Biology Department Travel award 2025 ($750)

Quantitative Life Sciences Travel Award 2024 *($500)*

Living Data Project Travel Award 2024 *($1800)*

CAMBAM Award ($1000)

Centre de recherches mathématiques general program grant 2023 ($4000)*

Biology department award ($1000)

FACTOR Showcase support - Musician's Travel Award ($500)

Deans Undergraduate Multidisciplinary Research list

*co-applied with Jake Harvey

RESEARCH MENTORSHIP

Marie Pyun - Undergraduate research project – *Linking modern coexistence theory to genome-scale models of bacteria (2025-2026)*

Jim Prescott - Undergraduate thesis project – *Fragmentation of dispersal networks affects synchrony in predator-prey model (2019)*

SKILLS

Technical Skills - Python, MATLAB, Julia, R, Java, C, F#

Lab Skills – Cloning, λ -Red recombineering, bacterial culture, microplate assays, PCR, genomic DNA extraction

Languages – English (native), Spanish (fluent), French (intermediate)